COMSATS University Islamabad, Lahore Campus

Terminal Examination – Fall 2024

Course Title:	Web Technologies			Course Code:	CSC336	Credit	4(3,1)
Course Instructor/s:	Asif Shahzad			Programme Name:	BSCS		
Semester:	6	Batch:	SP22	Section:	G1, G2	Date:	Jan. 08, 2024
Time Allowed:	**3 Hours**			**Maximum Marks:**	**50**		
Student's Name:				Reg. No.			
Note: Please answer only what is asked. Write very neat, keeping the font-size small. No marks for writing stuff that is not asked. You can skip the includes, middle-wares, html, head and body tags etc.							

Question 1 [CLO-1] Bloom Taxonomy Level: Understanding

Answer very briefly: (a). what is usage of cookies and sessions in web applications, give 2 examples for each, also enlist the headers involved in cookies creation and reading? (b). make a request-response flow diagram to explain how *redirect* works, what is role of *location* header in redirect? [5X2 = 10 Marks]

Question 2 [CLO-2] Bloom Taxonomy Level: Applying

Create an HTML page using Bootstrap with a fixed-width container, two rows, each having 2 cells. The first row cells should stacks vertically on medium and lower devices, while the second row cells should stacks only on extra small devices? [10 Marks]

Question 3 [CLO-3] Bloom Taxonomy Level: Applying

Create a webpage with a text input for Book ID and a "Load" button. The page should also have "Title" text input and "Availability" radio buttons (Available/Not Available). On clicking "Load," fetch book info via AJAX from /get-book endpoint, passing the Book ID as a query parameter. Assume, the response is a JSON object with title (string) and available (boolean). Auto-fill the Title and Availability fields based on the response. [15 Marks]

Question 4 [CLO-4] Bloom Taxonomy Level: Applying

Assume operations.js module export *getAllBooks* (that returns array of all book objects, available and not available both), *getBook* (that takes book ID and return the book object containing *title* and *available* fields) functions. Define Express GET endspoints i.e. /*display-available-books* that should return available books (ID and title) in tabular form using EJS/HTML and /*get-book* that should return book info as JSON whose ID is passed as query string. [15 Marks]